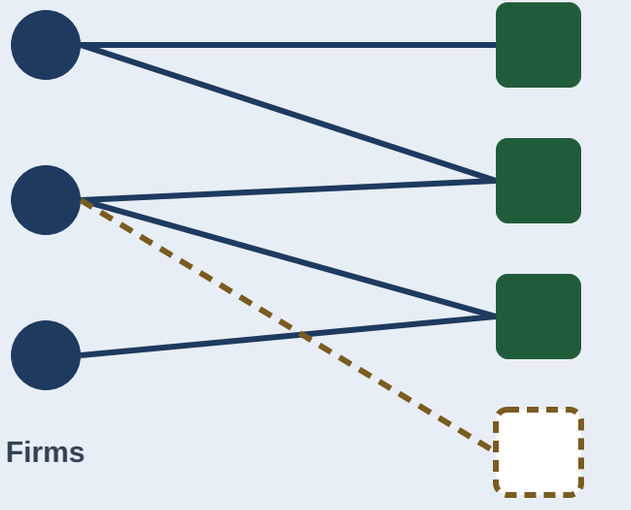
Time-valid graph G(t)



Predict first entry in t+1

Edges and features dated ≤ t

Embargoed holdout

2010-2012 burn-in
Freeze 30 CPC candidates

2013-2019 train
1,540,698 pairs

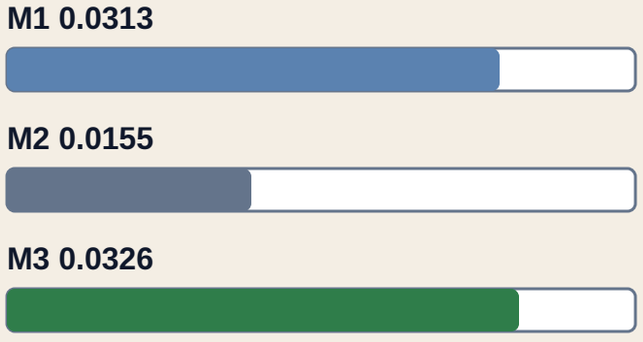
2020 embargo
Unused year

2021-2022 test
10,411 firms, 1.11% positives

M1, M2 activity; M3 + graph

CN, peer adoption, degree; no GNN

Held-out AP



M3 minus M2 forest

ΔAP +0.017

95% 0.0155 to 0.0191

AP 0.033 vs 1.11% base

Inspectable, not decision-ready